

Andrew D. Gordon

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Degrees

- BSc (Computer Science, First Class Honours) 1987, University of Edinburgh.
- PhD (Computer Science) 1992, University of Cambridge.

Professional Experience

- Research assistant, University of Edinburgh, summer 1987.
- Summer intern, Digital Systems Research Center, summer 1989.
- Research assistant, University of Cambridge, January 1991–October 1992.
- Forskarassistent, Chalmers University, January–December 1993.
- Research associate, University of Cambridge, January 1994–September 1994.
- Royal Society University Research Fellow, University of Cambridge, October 1994–October 1997.
- Partner Research Manager (and other roles), Microsoft Research, November 1997–November 2023.
- Visiting Professor at the University of Provence, Marseille, April 1998.
- Visiting Professor at the University of Newcastle, March 2007–March 2010.
- Professor of Computer Security, University of Edinburgh, October 2010–July 2023.
- Chief Science Officer, Cogna, November 2023–February 2025.
- Science Advisor, Evara AI, since March 2024.
- Science Advisor, Cogna, since March 2025.
- Science Advisor, Advanced Research + Innovation Agency (ARIA), since August 2025.

Awards and Honours

- Distinguished Dissertation in Computer Science, jointly awarded by the British Computer Society and the Conference of Professors and Heads of Computing, 1993.
- Most Influential ETAPS 1998 Paper, *Mobile Ambients*, with L. Cardelli, awarded by the European Association for Programming Languages and Systems, 2007.
- Most Influential POPL 2000 Paper, *Anytime, Anywhere: Modal Logics for Mobile Ambients*, with L. Cardelli, awarded by ACM SIGPLAN, 2010.
- Best Paper ETAPS 2013, *Deriving Probability Density Functions from Probabilistic Functional Programs*, with S. Bhat, J. Borgström, and C. Russo, awarded by the European Association for Programming Languages and Systems, 2013.
- ACM Fellow 2020, for “contributions to programming languages: their principles, logic, usability, and trustworthiness.” (ACM is the leading international association of computing professionals. ACM’s most prestigious member grade recognizes the top 1% of ACM members for their outstanding accomplishments in computing and information technology and/or outstanding service to ACM and the larger computing community.)
- Best paper, Honorable mention for ACM CHI 2023 paper, “*What It Wants Me To Say*”: *Bridging the Abstraction Gap Between End-User Programmers and Code-Generating Large Language Models*, with M. X. Liu, A. Sarkar, C. Negreanu, B. Zorn, J. Williams, and N. Toronto.
- Best paper, Honorable mention for IEEE VL/HCC 2023 paper, *FxD: a functional debugger for dysfunctional spreadsheets*, with I. Drosos, N. Wilson, S. Srinivasa Ragavan, and J. Williams.
- Honorary Professor, University of Edinburgh, since August 2023.
- Recipient of the 2024 *ETAPS Test of Time Award* with L. Cardelli for our FoS-SaCS 1998 paper *Mobile Ambients*.
- Recipient of the 2024 *Test of Time Award of the Symposium on Computer Security Foundations* with A. Jeffrey for our CSFW 2001 paper *Authenticity by Typing for Security Protocols*.

Publications

Journal Publications

1. M. Abadi and A. D. Gordon. A bisimulation method for cryptographic protocols. *Nordic Journal of Computing*, 5:267–303, 1998.

2. R. L. Crole and A. D. Gordon. Relating operational and denotational semantics for input/output effects. *Mathematical Structures in Computer Science*, 9:125–158, 1999.
3. M. Abadi and A. D. Gordon. A calculus for cryptographic protocols: The spi calculus. *Information and Computation*, 148:1–70, 1999.
4. A. D. Gordon. Bisimilarity as a theory of functional programming. *Theoretical Computer Science*, 228:5–47, 1999.
5. A. D. Gordon, S. B. Lassen, and P. D. Hankin. Compilation and equivalence of imperative objects. *Journal of Functional Programming*, 9(4):373–426, 1999.
6. L. Cardelli and A. D. Gordon. Mobile ambients. *Theoretical Computer Science*, 240:177–213, 2000.
7. L. Cardelli, G. Ghelli, and A. D. Gordon. Types for the ambient calculus. *Information and Computation*, 177:160–194, 2002.
8. A. D. Gordon and L. Cardelli. Equational properties of mobile ambients. *Mathematical Structures in Computer Science*, 12:1–38, 2002.
9. S. Dal Zilio and A. D. Gordon. Region analysis and a π -calculus with groups. *Journal of Functional Programming*, 12(3):229–292, 2002.
10. W. Charatonik, S. Dal Zilio, A. D. Gordon, S. Mukhopadhyay, and J.-M. Talbot. The complexity of model checking mobile ambients. *Theoretical Computer Science*, 308:277–331, 2003.
11. A. D. Gordon and A. Jeffrey. Typing correspondence assertions for communication protocols. *Theoretical Computer Science*, 300:379–409, 2003.
12. C. Fournet and A. D. Gordon. Stack inspection: Theory and variants. *ACM Transactions on Programming Languages and Systems (TOPLAS)*, 25(3):360–399, 2003.
13. A. D. Gordon and A. Jeffrey. Authenticity by typing for security protocols. *Journal of Computer Security*, 11(4):451–521, 2003.
14. A. D. Gordon and A. Jeffrey. Types and effects for asymmetric cryptographic protocols. *Journal of Computer Security*, 12(3/4):435–484, 2003.
15. K. Bhargavan, C. Fournet, and A. D. Gordon. A semantics for web services authentication. *Theoretical Computer Science*, 340(1):102–153, 2005.
16. A. D. Gordon and R. Pucella. Validating a web service security abstraction by typing. *Formal Aspects of Computing*, 17:277–318, 2005.
17. L. Cardelli, G. Ghelli, and A. D. Gordon. Secrecy and group creation. *Information and Computation*, 196(2):127–155, 2005.

18. C. Calcagno, L. Cardelli, and A. D. Gordon. Deciding validity in a spatial logic for trees. *Journal of Functional Programming*, 15:543–572, 2005.
19. L. Cardelli and A. D. Gordon. Ambient logic. *Mathematical Structures in Computer Science*. To appear.
20. K. Bhargavan, R. Corin, C. Fournet, and A. D. Gordon. Secure sessions for Web services. *ACM Transactions on Information and System Security*, 10(2), 2007.
21. C. Fournet, A. D. Gordon, and S. Maffei. A type discipline for authorization policies. *ACM Transactions on Programming Languages and Systems*, 29(5), 2007.
22. K. Bhargavan, C. Fournet, A. D. Gordon. Verifying policy-based web services security. *ACM Transactions on Programming Languages and Systems (TOPLAS)*, 30(6), 2008.
23. K. Bhargavan, C. Fournet, A.D. Gordon and S. Tse. Verified interoperable implementations of security protocols. *ACM Transactions on Programming Languages and Systems (TOPLAS)*, 31(1), 2008.
24. M. Becker, C. Fournet, and A. D. Gordon. SecPAL: Design and Semantics of a Decentralized Authorization Language. *Journal of Computer Security*, 18(4):597–643, 2010.
25. J. Bengtson, K. Bhargavan, C. Fournet, A. D. Gordon, and S. Maffei. Refinement types for secure implementations. *ACM Transactions on Programming Languages and Systems (TOPLAS)*, 33(2):8, 2011.
26. J. Borgström, A. D. Gordon, R. Pucella. Roles, stacks, histories: A triple for Hoare. *Journal of Functional Programming (JFP)*, 21(2):159–207, 2011.
27. G. M. Bierman, A. D. Gordon, C. Hrițcu, D. E. Langworthy. Semantic subtyping with an SMT solver. *Journal of Functional Programming (JFP)* 22(1):31–105, 2012.
28. J. Borgström, A. D. Gordon, M. Greenberg, J. Margetson, J. Van Gael. Measure Transformer Semantics for Bayesian Machine Learning. *Logical Methods in Computer Science* 9(3), 2013.
29. F. Dupressoir, A. D. Gordon, J. Jürjens, D. A. Naumann. Guiding a General-Purpose C Verifier to Prove Cryptographic Protocols. *Journal of Computer Security* 22(5):823–866, 2014.
30. V. Vaglica, M. Sajeve, H. N. McGough, D. Hutchison, C. Russo, A. D. Gordon, A. V. Ramarosandratana, W. Stuppy, M. J. Smith. Monitoring internet trade to inform species conservation actions. *Endangered Species Research* 32:223–235, 2017.

31. S. Bhat, J. Borgström, A. D. Gordon, and C. V. Russo. Deriving probability density functions from probabilistic functional programs. *Logical Methods in Computer Science*, 13(2), 2017.
32. M. McCutchen, J. Borghouts, A. D. Gordon, S. Peyton Jones, and A. Sarkar. Elastic sheet-defined functions: Generalising spreadsheet functions to variable-size input arrays. *J. Funct. Program.*, 30:e26, 2020.
33. M. I. Gorinova, A. D. Gordon, C. Sutton, and M. Vákár. Conditional independence by typing. *ACM Trans. Program. Lang. Syst.*, 44(1):4:1–4:54, 2022.
34. S. Chen, A. Karaoglu, C. Negreanu, T. Ma, J.-G. Yao, J. Williams, F. Jiang, A. D. Gordon, and C.-Y. Lin. LinkingPark: An automatic semantic table interpretation system. *J. Web Semant.*, 74:100733, 2022.
35. D. Robinson, C. Cabrera, A. D. Gordon, N. D. Lawrence, L. Mennen. Requirements Are All You Need: The Final Frontier for End-User Software Engineering *ACM Transactions on Software Engineering and Methodology*, 34(5):1-22, 2025.

Refereed Conference and Workshop Publications

1. A. D. Gordon. The formal definition of a synchronous hardware-description language in higher order logic. In *International Conference on Computer Design, Cambridge, Massachusetts, October 11–14, 1992*, pages 531–534. IEEE Computer Society Press, 1992.
2. A. D. Gordon. An operational semantics for I/O in a lazy functional language. In *FPCA’93: Conference on Functional Programming Languages and Computer Architecture, Copenhagen*, pages 136–145. ACM Press, 1993.
3. R. L. Crole and A. D. Gordon. Factoring an adequacy proof (preliminary report). In *Functional Programming, Glasgow 1993*, Workshops in Computing, pages 9–27. Springer-Verlag, 1994.
4. A. D. Gordon. A mechanisation of name-carrying syntax up to alpha-conversion. In J. J. Joyce and C.-J. H. Seger, editors, *Higher Order Logic Theorem Proving and its Applications. Proceedings, 1993*, number 780 in Lecture Notes in Computer Science, pages 414–426. Springer-Verlag, 1994.
5. R. L. Crole and A. D. Gordon. A sound metalogical semantics for input/output effects. In L. Pacholski and J. Tiuryn, editors, *CSL’94 Computer Science Logic, Kazimierz, Poland, September 1994*, volume 933 of *Lecture Notes in Computer Science*, pages 339–353. Springer-Verlag, 1995.
6. A. D. Gordon. A tutorial on co-induction and functional programming. In *Functional Programming, Glasgow 1994*, Workshops in Computing, pages 78–95. Springer-Verlag, 1995.

7. A. D. Gordon. Bisimilarity as a theory of functional programming. In *Eleventh Annual Conference on Mathematical Foundations of Programming Semantics*, volume 1 of *Electronic Notes in Theoretical Computer Science*. Elsevier Science Publishers B.V., 1995.
8. A. D. Gordon and K. Hammond. Monadic I/O in Haskell 1.3. In Paul Hudak, editor, *Proceedings of the Haskell Workshop, June 25, 1995, La Jolla, California*, pages 50–68, 1995. Available as Yale University Research Report YALEU/DCS/RR–1075.
9. A. D. Gordon and G. D. Rees. Bisimilarity for a first-order calculus of objects with subtyping. In *23rd ACM Symposium on Principles of Programming Languages (POPL’96)*, pages 386–395. ACM Press, 1996.
10. S. L. Peyton Jones, A. D. Gordon, and S. Finne. Concurrent Haskell. In *23rd ACM Symposium on Principles of Programming Languages (POPL’96)*, pages 295–308. ACM Press, 1996.
11. A. D. Gordon and T. Melham. Five axioms of alpha-conversion. In *Theorem Proving in Higher Order Logics: 9th International Conference, TPHOLs’96*, volume 1125 of *Lecture Notes in Computer Science*, pages 173–191. Springer-Verlag, 1996.
12. M. Abadi and A. D. Gordon. A calculus for cryptographic protocols: The spi calculus. In *4th ACM Conference on Computer and Communications Security (CCS’97)*, pages 36–47. ACM Press, April 1997.
13. M. Abadi and A. D. Gordon. Reasoning about cryptographic protocols in the spi calculus. In *Concurrency Theory (CONCUR’97)*, Lecture Notes in Computer Science, pages 59–73. Springer-Verlag, August 1997.
14. A. D. Gordon, S. B. Lassen, and P. D. Hankin. Compilation and equivalence of imperative objects. In *Foundations of Software Technology and Theoretical Computer Science (FST&TCS’97)*, volume 1346 of *Lecture Notes in Computer Science*, pages 74–87. Springer-Verlag, 1997.
15. A. D. Gordon. Operational equivalences for untyped and polymorphic object calculi. In A. D. Gordon and A. M. Pitts, editors, *Higher Order Operational Techniques in Semantics*, Publications of the Newton Institute, pages 9–54. Cambridge University Press, 1998.
16. M. Abadi and A. D. Gordon. A bisimulation method for cryptographic protocols. In *European Symposium on Programming (ESOP’98)*, volume 1381 of *Lecture Notes in Computer Science*, pages 12–26. Springer-Verlag, 1998.
17. L. Cardelli and A. D. Gordon. Mobile ambients. In *Foundations of Software Science and Computation Structures (FOSSACS’98)*, volume 1378 of *Lecture Notes in Computer Science*, pages 140–155. Springer-Verlag, 1998.

18. A. D. Gordon and P. D. Hankin. A concurrent object calculus: reduction and typing. In *3rd International Workshop on High-Level Concurrent Languages (HLCL'98)*, volume 16 of *Electronic Notes in Theoretical Computer Science*. Elsevier, 1998.
19. L. Cardelli and A. D. Gordon. Types for mobile ambients. In *26th ACM Symposium on Principles of Programming Languages (POPL'99)*, pages 79–92. ACM Press, 1999.
20. L. Cardelli, G. Ghelli, and A. D. Gordon. Mobility types for mobile ambients. In *International Conference on Automata, Languages, and Programming (ICALP'99)*, volume 1644 of *Lecture Notes in Computer Science*, pages 230–239. Springer-Verlag, 1999.
21. A. D. Gordon and L. Cardelli. Equational properties of mobile ambients. In *Foundations of Software Science and Computation Structures*, Lecture Notes in Computer Science, pages 212–226. Springer-Verlag, 1999.
22. L. Cardelli and A. D. Gordon. Anytime, anywhere: Modal logics for mobile ambients. In *27th ACM Symposium on Principles of Programming Languages (POPL'00)*, pages 365–377. ACM Press, 2000.
23. L. Cardelli, G. Ghelli, and A. D. Gordon. Ambient groups and mobility types. In *Proceedings TCS2000*, volume 1872 of *Lecture Notes in Computer Science*, pages 333–347. Springer-Verlag, 2000.
24. L. Cardelli, G. Ghelli, and A. D. Gordon. Secrecy and group creation. In *Concurrency Theory (CONCUR'00)*, volume 1877 of *Lecture Notes in Computer Science*, pages 365–379. Springer-Verlag, 2000.
25. A. D. Gordon and D. Syme. Typing a multi-language intermediate code. In *28th ACM Symposium on Principles of Programming Languages (POPL'01)*, pages 248–260, 2001.
26. W. Charatonik, S. Dal Zilio, A. D. Gordon, S. Mukhopadhyay, and J.-M. Talbot. The complexity of model checking mobile ambients. In *Foundations of Software Science and Computation Structures (FOSSACS'01)*, volume 2030 of *Lecture Notes in Computer Science*, pages 152–167. Springer-Verlag, 2001.
27. L. Cardelli and A. D. Gordon. Logical properties of name restriction. In *Typed Lambda Calculi and Applications (TLCA'01)*, volume 2044 of *Lecture Notes in Computer Science*, pages 46–60. Springer-Verlag, 2001.
28. A. D. Gordon and A. Jeffrey. Typing correspondence assertions for communication protocols. In *Mathematical Foundations of Programming Semantics 17*, *Electronic Notes in Theoretical Computer Science*. Elsevier, 2001. Pages 99–120 of the Preliminary Proceedings, BRICS Notes Series NS-01-2, BRICS, University of Aarhus, May 2001.

29. A. D. Gordon and A. Jeffrey. Authenticity by typing for security protocols. In *14th IEEE Computer Security Foundations Workshop*, pages 145–159. IEEE Computer Society Press, 2001.
30. C. Fournet and A. D. Gordon. Stack inspection: Theory and variants. In *29th ACM Symposium on Principles of Programming Languages (POPL'02)*, pages 307–318, 2002.
31. W. Charatonik, A. D. Gordon, and J.-M. Talbot. Finite-control mobile ambients. In *European Symposium on Programming (ESOP'02)*, volume 2305 of *Lecture Notes in Computer Science*, pages 295–313. Springer-Verlag, 2002.
32. A. D. Gordon and A. Jeffrey. Types and effects for asymmetric cryptographic protocols. In *15th IEEE Computer Security Foundations Workshop*, pages 77–91. IEEE Computer Society Press, 2002.
33. D. Syme and A. D. Gordon. Automating type soundness proofs via decision procedures and guided reductions. In *9th International Conference on Logic for Programming Artificial Intelligence and Reasoning*, volume 2514 of *Lecture Notes in Computer Science*, pages 418–434. Springer-Verlag, 2002.
34. A. D. Gordon and R. Pucella. Validating a web service security abstraction by typing. In *2002 ACM Workshop on XML Security*, pages 18–29, 2002.
35. A. D. Gordon and A. Jeffrey. Typing one-to-one and one-to-many correspondences in security protocols. In *Software Security - Theories and Systems (ISSS 2002)*, volume 2609 of *Lecture Notes in Computer Science*, pages 263–282. Springer-Verlag, 2002.
36. C. Calcagno, L. Cardelli, and A. D. Gordon. Deciding validity in a spatial logic for trees. In *ACM SIGPLAN Workshop on Types in Language Design and Implementation (TLDI)*, pages 62–73, 2003.
37. K. Bhargavan, C. Fournet, and A. D. Gordon. A semantics for web services authentication. In *31st ACM Symposium on Principles of Programming Languages (POPL'04)*, pages 198–209, 2004.
38. K. Bhargavan, C. Fournet, and A. D. Gordon. Verifying policy-based security for web services. In *11th ACM Conference on Computer and Communications Security (CCS'04)*, pages 268–277, October 2004.
39. K. Bhargavan, R. Corin, C. Fournet, and A. D. Gordon. Secure sessions for web services. In *2004 ACM Workshop on Secure Web Services (SWS'04)*, pages 56–66, 2004.
40. C. Fournet, A. D. Gordon, and S. Maffei. A type discipline for authorization policies. In *European Symposium on Programming (ESOP'05)*, volume 3444 of *Lecture Notes in Computer Science*, pages 141–156. Springer-Verlag, 2005.

41. A. D. Gordon and A. Jeffrey. Secrecy despite compromise: Types, cryptography, and the pi-calculus. In *Concurrency Theory (CONCUR'05)*, volume 3653 of *Lecture Notes in Computer Science*, pages 186–201. Springer-Verlag, 2005.
42. K. Bhargavan, C. Fournet, A. D. Gordon, and G. O'Shea. An advisor for web services security policies. In *2005 ACM Workshop on Secure Web Services*, pages 1–9. ACM, 2005.
43. K. Bhargavan, C. Fournet, A. D. Gordon, and S. Tse. Verified interoperable implementations of security protocols. In *19th IEEE Computer Security Foundations Workshop (CSFW'06)*, pages 139–152, 2006.
44. K. Bhargavan, C. Fournet, and A. D. Gordon. Verified reference implementations of WS-Security protocols. In *3rd International Workshop on Web Services and Formal Methods (WS-FM 2006)*, volume 4184 of *Lecture Notes in Computer Science*, pages 88–106. Springer-Verlag, 2006.
45. M. Becker, C. Fournet, and A. D. Gordon. SecPAL: Design and Semantics of a Decentralized Authorization Language. In *20th IEEE Symposium on Computer Security Foundations (CSF'07)*, pages 3-15, Venice, July 6-8, 2007.
46. C. Fournet, A. D. Gordon, and S. Maffei. A type discipline for authorization in distributed systems. In *20th IEEE Symposium on Computer Security Foundations (CSF'07)*, pages 31-48, Venice, July 6-8, 2007. IEEE Computer Society.
47. J. Borgström, A. D. Gordon, and A. Phillips. A chart semantics for the pi-calculus. In *14th International Workshop on Expressiveness in Concurrency (EXPRESS'07)*, Electronic Notes in Theoretical Computer Science 194(2):3-29 (January 2008).
48. K. Bhargavan, C. Fournet, A. D. Gordon, and N. Swamy. Verified implementations of the information card federated identity-management protocol. In *ACM Symposium on Information, Computer and Communication Security (ASIACCS '08)*, Tokyo, March 18-20, 2008.
49. K. Bhargavan, A. D. Gordon, and I. Narasamdya. Service combinators for farming virtual machines. In *COORDINATION 2008*, Oslo, June 4-6, 2008. Springer LNCS 5052:33-49.
50. J. Bengtson, K. Bhargavan, C. Fournet, A. D. Gordon, and S. Maffei. Refinement types for secure implementations. In *21st IEEE Computer Security Foundations Symposium (CSF 2008)*, pages 17-32, Pittsburgh, Pennsylvania, June 23-25, 2008. IEEE Computer Society.
51. A. D. Gordon, H. Hüttel, and R. Rydhof Hansen. Type inference for correspondence types. In *6th International Workshop on Security Issues in Concurrency (SecCo 2008)*, Toronto, August 23, 2008.

52. S. Maffei, M. Abadi, C. Fournet, and A. D. Gordon. Code-carrying authorization. In *13th European Symposium on Research in Computer Security (ESORICS 2008)*, Malaga, October 6-8, 2008. Springer LNCS 5283:563-579.
53. I. Baltopoulos and A. D. Gordon. Secure compilation of a multi-tier web language. In *ACM SIGPLAN Workshop on Types in Language Design and Implementation (TLDI 2009)*, Savannah, Georgia, January 24, 2009. ACM Press. Pages 27-38.
54. A. Mukhamedov, A. D. Gordon, and M. Ryan. Towards a verified reference implementation of a trusted platform module. In *Seventeenth International Workshop on Security Protocols*, Cambridge, April 1-3, 2009. Springer LNCS 7028:69-81.
55. J. Borgström, K. Bhargavan, and A. D. Gordon. A compositional theory for STM Haskell. In *ACM SIGPLAN Haskell Symposium*, Edinburgh, September 3, 2009. Pages 69-80. ACM Press.
56. K. Bhargavan, C. Fournet, and A. D. Gordon. Modular verification of security protocol code by typing. In *37th ACM Symposium on Principles of Programming Languages (POPL'10)*, pages 198-209, 2010.
57. F. Dupressoir, A. D. Gordon, and J. Jürjens. Verifying authentication properties of C security protocol code using general verifiers. In *4th International Workshop on Analysis of Security APIs (ASA-4)*, 2010.
58. G. M. Bierman, A. D. Gordon, C. Hrițcu and D. Langworthy. Semantic subtyping with an SMT solver. In *International Conference on Functional Programming (ICFP'10)*, pages 105-116, Baltimore, September 2010. ACM Press.
59. J. Borgström, A. D. Gordon, M. Greenberg, J. Margetson, J. Van Gael. Measure Transformer Semantics for Bayesian Machine Learning. *European Symposium on Programming (ESOP'11)*, pages 77-96.
60. I. G. Baltopoulos, J. Borgström, A. D. Gordon. Maintaining Database Integrity with Refinement Types. *European Conference on Object Oriented Programming (ECOOP 2011)*, pages 484-509.
61. F. Dupressoir, A. D. Gordon, J. Jürjens, D. A. Naumann. Guiding a General-Purpose C Verifier to Prove Cryptographic Protocols. *IEEE Symposium on Computer Security Foundations (CSF 2011)*, pages 3-17.
62. M. Aizatulin, A. D. Gordon, J. Jürjens. Extracting and verifying cryptographic models from C protocol code by symbolic execution. *ACM Conference on Computer and Communications Security (CCS 2011)*, pages 331-340.
63. M. Aizatulin, A. D. Gordon, J. Jürjens. Computational verification of C protocol implementations by symbolic execution. *ACM Conference on Computer and Communications Security (CCS 2012)*, pages 712-723.

64. J. A. Hewson, P. Anderson, A. D. Gordon. A Declarative Approach to Automated Configuration. *Proceedings of the 26th Large Installation System Administration Conference (LISA 2012)*, pages 51–66.
65. A. D. Gordon, M. Aizatulin, J. Borgström, G. Claret, T. Graepel, A. V. Nori, S. K. Rajamani, C. V. Russo. A model-learner pattern for Bayesian reasoning. In *40th ACM Symposium on Principles of Programming Languages (POPL'13)*, pages 403–416, 2013.
66. S. Bhat, J. Borgström, A. D. Gordon, C. V. Russo. Deriving Probability Density Functions from Probabilistic Functional Programs. In *Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2013)*, pages 508–522, 2013.
67. G. Claret, S. K. Rajamani, A. V. Nori, A. D. Gordon, J. Borgström: Bayesian inference using data flow analysis. In *ACM SIGSOFT Symposium on the Foundations of Software Engineering (FSE 2013)*, pages 92–102.
68. J. A. Hewson, P. Anderson, A. D. Gordon. Constraint-Based Autonomic Reconfiguration. *7th IEEE International Conference on Self-Adaptive and Self-Organizing Systems (SASO 2013)*, pages 101–110, 2013.
69. A. D. Gordon, T. Graepel, N. Rolland, C. V. Russo, J. Borgström, J. Guiver. Tabular: a schema-driven probabilistic programming language. In *41st ACM Symposium on Principles of Programming Languages (POPL'14)*, pages 321–334, 2014.
70. A. D. Gordon, C. V. Russo, M. Szymczak, J. Borgström, N. Rolland, T. Graepel, D. Tarlow. Probabilistic Programs as Spreadsheet Queries. In *European Symposium on Programming (ESOP'15)*, pages 1–25, 2015.
71. M. Allamanis, D. Tarlow, A.D. Gordon, Yi Wei. Bimodal Modelling of Source Code and Natural Language. In *International Conference on Machine Learning (ICML 2015)*, pages 2123–2132, 2015.
72. A. Scibior, Z. Ghahramani, A.D. Gordon. Practical probabilistic programming with monads. In *ACM Haskell Symposium (Haskell 2015)*, pages 165–176, 2015.
73. J. Borgström, A. D. Gordon, L. Ouyang, C.V. Russo, A. Scibior, M. Szymczak. Fabular: regression formulas as probabilistic programming. In *43rd ACM Symposium on Principles of Programming Languages (POPL'16)*, pages 271–283, 2016.
74. W. Chen, D. Aspinall, A. D. Gordon, C.A. Sutton, I. Muttik. Explaining unwanted behaviours in context. In *1st International Workshop on Innovations in Mobile Privacy and Security (IMPS 2016)*, 2016.
75. W. Chen, D. Aspinall, A. D. Gordon, C.A. Sutton, I. Muttik. A text-mining approach to explain unwanted behaviours. In *2016 European Workshop on System Security (EUROSEC 2016)*.

76. W. Chen, D. Aspinall, A. D. Gordon, C.A. Sutton, I. Muttik. On robust malware classifiers by verifying unwanted behaviours. In *Integrated Formal Methods - 12th International Conference, IFM 2016*, pages 326–341.
77. W. Chen, D. Aspinall, A. D. Gordon, C.A. Sutton, I. Muttik. More semantics more robust: improving Android malware classifiers. In *Proceedings of the 9th ACM Conference on Security & Privacy in Wireless and Mobile Networks, WISEC 2016*, pages 147–158.
78. G. Barthe, G. P. Farina, M. Gaboardi, E. J. Gallego Arias, A. D. Gordon, J. Hsu, P.-Y. Strub. Differentially private Bayesian programming. In *23rd ACM Conference on Computer and Communications Security (CCS'16)*, pages 68–79.
79. J. Borgström, U. Dal Lago, A. D. Gordon, and M. Szymczak. A lambda-calculus foundation for universal probabilistic programming. In *International Conference on Functional Programming (ICFP'16)*, pages 33–46. ACM Press.
80. A. Sarkar, A. D. Gordon, S. Peyton Jones, and N. Toronto. Calculation view: multiple-representation editing in spreadsheets. In *2018 IEEE Symposium on Visual Languages and Human-Centric Computing, VL/HCC 2018, Lisbon, Portugal, October 1-4, 2018*, pages 85–93. IEEE Computer Society, 2018.
81. A. Sarkar and A. D. Gordon. How do people learn to use spreadsheets? (Work in progress). In *Proceedings of the 29th Annual Workshop of the Psychology of Programming Interest Group, PPIG 2018, London, UK, September 5 - 7, 2018*.
82. M. I. Gorinova, A. D. Gordon, and C. A. Sutton. Probabilistic programming with densities in SlicStan: efficient, flexible, and deterministic. *Proc. ACM Program. Lang.*, 3(POPL):35:1–35:30, 2019.
83. A. F. Blackwell, L. Church, M. Erwig, J. Geddes, A. D. Gordon, M. I. Gorinova, A. G. Baydin, B. Gram-Hansen, T. Kohn, N. Lawrence, V. Mansinghka, B. Paige, T. Petricek, D. Robinson, A. Sarkar, O. Strickson. In *Proceedings of the 30th Annual Workshop of the Psychology of Programming Interest Group, PPIG 2019, Newcastle University, UK, August 28 - 30, 2019*.
84. N. Joharizadeh, A. Sarkar, A. D. Gordon, and J. Williams. Gridlets: Reusing spreadsheet grids. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems, CHI 2020, Honolulu, HI, USA, April 25-30, 2020*, pages 1–7. ACM, 2020.
85. J. Williams, N. Joharizadeh, A. D. Gordon, and A. Sarkar. Higher-order spreadsheets with spilled arrays. In *Programming Languages and Systems - 29th European Symposium on Programming, ESOP 2020, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2020, Dublin, Ireland, April 25-30, 2020, Proceedings*, volume 12075 of *Lecture Notes in Computer Science*, pages 743–769. Springer, 2020.

86. S. Chen, A. Karaoglu, C. Negreanu, T. Ma, J.-G. Yao, J. Williams, A. D. Gordon, and C.-Y. Lin. Linkingpark: An integrated approach for semantic table interpretation. In *Proceedings of the Semantic Web Challenge on Tabular Data to Knowledge Graph Matching (SemTab 2020) co-located with the 19th International Semantic Web Conference (ISWC 2020), Virtual conference (originally planned to be in Athens, Greece), November 5, 2020*, volume 2775 of *CEUR Workshop Proceedings*, pages 65–74. CEUR-WS.org, 2020.
87. J. Williams, C. Negreanu, A. D. Gordon, and A. Sarkar. Understanding and inferring units in spreadsheets. In *IEEE Symposium on Visual Languages and Human-Centric Computing, VL/HCC 2020, Dunedin, New Zealand, August 10-14, 2020*, pages 1–9. IEEE, 2020.
88. S. Srinivasa Ragavan, A. Sarkar, and A. D. Gordon. Spreadsheet comprehension: Guesswork, giving up and going back to the author. In *CHI '21: CHI Conference on Human Factors in Computing Systems, Virtual Event / Yokohama, Japan, May 8-13, 2021*, pages 181:1–181:21. ACM, 2021.
89. J. Williams and A. D. Gordon. Where-provenance for bidirectional editing in spreadsheets. In *IEEE Symposium on Visual Languages and Human-Centric Computing, VL/HCC 2021, St Louis, MO, USA, October 10-13, 2021*, pages 1–10. IEEE, 2021.
90. C. Negreanu, A. Karaoglu, J. Williams, S. Chen, D. Fabian, A. D. Gordon, and C.-Y. Lin. Rows from many sources: Enriching row completions from wikidata with a pre-trained language model. In *Companion of The Web Conference 2022, Virtual Event / Lyon, France, April 25 - 29, 2022*, pages 1272–1280. ACM, 2022.
91. S. Srinivasa Ragavan, Z. Hou, Y. Wang, A. D. Gordon, H. Zhang, and D. Zhang. Gridbook: Natural language formulas for the spreadsheet grid. In *IUI 2022: 27th International Conference on Intelligent User Interfaces, Helsinki, Finland, March 22 - 25, 2022*, pages 345–368. ACM, 2022.
92. A. Sarkar, A. D. Gordon, C. Negreanu, C. Poelitz, S. Srinivasa Ragavan, and B. Zorn. What is it like to program with artificial intelligence? In *Proceedings of the 33rd Annual Conference of the Psychology of Programming Interest Group (PPIG 2022)*, 2022. Available at <https://arxiv.org/abs/2208.06213>.
93. M. X. Liu, A. Sarkar, C. Negreanu, B. G. Zorn, J. Williams, N. Toronto, A. D. Gordon. “What It Wants Me To Say”: Bridging the Abstraction Gap Between End-User Programmers and Code-Generating Large Language Models. In *CHI 2023: Conference on Human Factors in Computing Systems, Hamburg, Germany, April 23-28, 2023*, pages 598:1–598:31. ACM, 2023.
94. I. Drosos, N. Wilson, S. Srinivasa Ragavan, J. Williams, A. D. Gordon. FxD: a functional debugger for dysfunctional spreadsheets. In *IEEE Symposium on Visual Languages and Human-Centric Computing, VL/HCC 2023, Washington, DC, USA, October 2-6, 2023*. IEEE, 2023.

95. K. Ferdowsi, J. Williams, I. Drosos, A. D. Gordon, C. Negreanu, N. Polikarpova, A. Sarkar, B. Zorn. ColDeco: An End User Spreadsheet Inspection Tool for AI-Generated Code. In *IEEE Symposium on Visual Languages and Human-Centric Computing, VL/HCC 2023, Washington, DC, USA, October 2-6, 2023*. IEEE, 2023.
96. A. D. Gordon (editor), C. Negreanu (editor), J. Cambronero, R. Chakravarthy, I. Drosos, H. Fang, B. Mitra, H. Richardson, A. Sarkar, S. Simmons, J. Williams, B. Zorn. Co-audit: tools to help humans double-check AI-generated content. In *PLATEAU 2024*, University of California Berkeley, February 2024. Available at https://kilthub.cmu.edu/articles/conference_contribution/Co-audit_tools_to_help_humans_double-check_AI-generated_content/25587552.
97. X. Sun, A. D. Gordon, H. Ge. Multi-Agent Systems for Traceable Bayesian Workflow. In *ACM Workshop on Languages for Inference (LAFI 2026)*, Rennes, January 2026. Available at <https://popl26.sigplan.org/details/lafi-2026-papers/16/Multi-Agent-Systems-for-Traceable-Bayesian-Workflow>.

Preprints

1. Z. Garby, A. D. Gordon, and D. Sands. The LLMbda Calculus: AI Agents, Conversations, and Information Flow. arXiv:2602.20064, February 2026 (revised July 2026). Available at <https://arxiv.org/abs/2602.20064>.
2. X. Sun, A. D. Gordon, and H. Ge. Agentic Bayesian Data Analysis: From Scientific Questions to Validated Probabilistic Models. Submitted for publication, May 2026.
3. G. Granberry, W. Ahrendt, M. Johansson, and A. D. Gordon. Bug Finding with Agentic Symbolic Execution. Submitted for publication, June 2026.

Invited Conference Publications

1. R. Boulton, A. Gordon, M. Gordon, J. Harrison, J. Herbert, and J. Van Tassel. Experience with embedding hardware description languages in HOL. In V. Stavridou, T. F. Melham, and R. T. Boute, editors, *Theorem Provers in Circuit Design: Theory, Practice and Experience: Proceedings of the IFIP TC10/WG 10.2 International Conference, Nijmegen, June 1992*, IFIP Transactions A-10, pages 129–156. North-Holland, 1992.
2. A. D. Gordon. Nominal calculi for security and mobility. In *DARPA Workshop on Foundations for Secure Mobile Code*, March 1997.
3. S. Dal Zilio and A. D. Gordon. Region analysis and a π -calculus with groups. In *Proceedings MFCS'00*, volume 1893 of *Lecture Notes in Computer Science*, pages 1–20. Springer-Verlag, 2000.

4. K. Bhargavan, C. Fournet, A. D. Gordon, and R. Pucella. TulaFale: A security tool for web services. In *International Symposium on Formal Methods for Components and Objects (FMCO'03)*, volume 3188 of *Lecture Notes in Computer Science*, pages 197–222. Springer-Verlag, 2004.
5. A. D. Gordon. Provable implementations of security protocols. In *Proceedings of the 21st IEEE Symposium on Logic in Computer Science (LICS'06)*, pages 345–346, 2006.
6. A. D. Gordon, R. Harper, J. Harrison, A. Jeffrey, P. Sewell. Robin Milner 1934–2010: verification, languages, and concurrency. *POPL* 2011:473–474.
7. M. Aizatulin, F. Dupressoir, A. D. Gordon, J. Jürjens. Verifying Cryptographic Code in C: Some Experience and the Csec Challenge. *Formal Aspects in Security and Trust* 2011:1–20.
8. A. D. Gordon, T. A. Henzinger, A. V. Nori, S. K. Rajamani. Probabilistic Programming. *Foundations of Software Engineering* 2014:167–181.
9. J. Borghouts, A. D. Gordon, A. Sarkar, and N. Toronto. End-user probabilistic programming. In David Parker and Verena Wolf, editors, *Quantitative Evaluation of Systems, 16th International Conference, QEST 2019, Glasgow, UK, September 10–12, 2019, Proceedings*, volume 11785 of *Lecture Notes in Computer Science*, pages 3–24. Springer, 2019.

Invited Book Chapters

1. A. D. Gordon. Example: the binomial theorem. In M. J. C. Gordon and T. F. Melham, editors, *Introduction to HOL: A theorem proving environment for higher order logic*, chapter 7, pages 77–95. Cambridge University Press, 1993.
2. A. D. Gordon. Notes on nominal calculi for security and mobility. In R. Focardi and R. Gorrieri, editors, *Foundations of Security Analysis and Design*, Lecture Notes in Computer Science, pages 262–330. Springer-Verlag, 2001.
3. K. Bhargavan, C. Fournet, and A. D. Gordon. Policy advisor for WSE 3.0. In *Web Service Security: Scenarios, patterns, and implementation guidance for Web Services Enhancements (WSE) 3.0*, pages 324–330. Microsoft Press, 2006.
4. K. Bhargavan, C. Fournet, A. D. Gordon, and S. Tse. Verified interoperable implementations of security protocols. In M. Broy, J. Grünbauer, C. A. R. Hoare (editors), *Software system reliability and security: Proceedings of the NATO Summer School Marktoberdorf 2006*, pages 87–115. IOS Press, 2007.
5. A. D. Gordon. Foreword. In P. Periorellis (editor), *Securing Web Services: Practical Usage of Standards and Specifications*, pages ix–x. Idea Group Inc (IGI), 2007.

6. A. D. Gordon and C. Fournet. Principles and applications of refinement types. In J. Esparza, B. Spanfelner, O. Grumberg (editors), *Logics and Languages for Reliability and Security: Proceedings of the NATO Summer School Marktoberdorf 2009*, pages 73–104. IOS Press, 2010. A preliminary version appears as Technical Report MSR-TR-2009-147, Microsoft Research, October 2009.
7. J. Borgström, A. D. Gordon, and R. Pucella. Roles, Stacks, Histories: A Triple for Hoare. To appear in a Festschrift volume for Tony Hoare, published by Springer. An extended version of this paper appears as Technical Report MSR-TR-2009-97, Microsoft Research, November 2009.
8. C. Fournet, K. Bhargavan, A. D. Gordon. Cryptographic Verification by Typing for a Sample Protocol Implementation. FOSAD 2011:66-100.
9. A. D. Gordon, C. Russo, M. Szymczak, J. Borgström, N. Rolland, T. Graepel, D. Tarlow. Tabular: Probabilistic Inference from the Spreadsheet. *Foundations of Probabilistic Programming*, MIT Press, 2020.

Books and Collections

1. A. D. Gordon. *Functional Programming and Input/Output*. Distinguished Dissertations in Computer Science. Cambridge University Press, 1994. Paperback edition, 2008.
2. A. D. Gordon and A. M. Pitts, editors. *Higher Order Operational Techniques in Semantics*, Publications of the Newton Institute. Cambridge University Press, 1998.
3. A. D. Gordon, A. M. Pitts, and C. L. Talcott, editors. *Higher Order Operational Techniques in Semantics*, volume 10 of *Electronic Notes in Theoretical Computer Science*. Elsevier, 1998.
4. A. D. Gordon and A. M. Pitts, editors. *Higher Order Operational Techniques in Semantics*, volume 26 of *Electronic Notes in Theoretical Computer Science*. Elsevier, 1999.
5. A. D. Gordon, editor. *Foundations of Software Science and Computation Structures (FOSSACS'03)*, volume 2620 of *Lecture Notes in Computer Science*. Springer-Verlag, 2003.
6. A. D. Gordon, editor. *Theoretical Computer Science*, 333(1–2):1–327, March 2005. Special issue based on FOSSACS'03.
7. L. Aceto and A. D. Gordon, editors. *Algebraic Process Calculi: The First Twenty Five Years and Beyond*, 2005. Short Contributions from the Workshop on Algebraic Process Calculi: The First Twenty Five Years and Beyond (PA-05), Bertinoro, Forlì, Italy, August 1-5, 2005. BRICS Note NS-05-03. Also appears as Volume 162, pages 1–340 of *Electronic Notes in Theoretical Computer Science*, Elsevier.

8. A. D. Gordon and D. Sands, editors. *Proceedings of the 2006 ACM Workshop on Formal Methods in Security Engineering (FMSE'06)*. ACM Press, 2006.
9. L. Aceto, M. Bravetti, W. Fokkink, and A. D. Gordon, editors. *Journal of Logic and Algebraic Programming*, 70(2):119–238, 2007. First special issue on *Algebraic Process Calculi: The First Twenty Five Years and Beyond*.
10. L. Aceto, M. Bravetti, W. Fokkink, and A. D. Gordon, editors. *Journal of Logic and Algebraic Programming*, 72(1):1–122, 2007. Second special issue on *Algebraic Process Calculi: The First Twenty Five Years and Beyond*.
11. L. Aceto, M. Bravetti, W. Fokkink, and A. D. Gordon, editors. *Journal of Logic and Algebraic Programming*, 75(1):1–166, 2008. Third special issue on *Algebraic Process Calculi: The First Twenty Five Years and Beyond*.
12. K. Bhargavan, A. D. Gordon, T. Harris, and P. Toft, editors. *Proceedings of the Joint MSR–HP Workshop on “The Rise and Rise of the Declarative Datacentre”*. Technical Report MSR–TR–2008–61, Microsoft Research. May 2008.
13. M. Abadi, P. Gardner, A. D. Gordon, and R. Mardare, editors. *Essays for the Luca Cardelli Fest*. Technical Report MSR–TR–2014–104, Microsoft Research. September 2014.
14. G. Barthe, A. D. Gordon, J.-P. Katoen, A. McIver, editors. *Challenges and Trends in Probabilistic Programming (Dagstuhl Seminar 15181)*. Dagstuhl Reports 5(4): 123–141, 2015.
15. G. Castagna, A. D. Gordon, editors. *Proceedings of the 44th ACM SIGPLAN Symposium on Principles of Programming Languages, POPL 2017, Paris, France, January 18–20, 2017*. ACM 2017.

Dissertation

1. A. D. Gordon. *Functional Programming and Input/Output*. PhD Dissertation, Cambridge University, 1992.

Research Community

Professional Service

- Served on Haskell 1.3 and 1.4 committees, to standardise I/O, 1993–97.
- Member of Steering Committee of *ICFP* 1999–2002, and *ETAPS*, 2002–2004, and since 2008.
- Elected member of UK EPSRC Peer Review College, 2000–2013.
- Member of Advisory Board of EC *MyTHS* project, 2002–2005.

- Founding member of Editorial Board of *Logical Methods in Computer Science*, 2004–2015.
- Member of IFIP WG1.8 on *Concurrency Theory*, 2005–2011.
- Member of Scientific Advisory Board of the *Excellence Cluster on Multimodal Computing and Interaction (M2CI)*, Saarbruecken, 2008–2016.
- Member of Selection Committee for the *ACM SIGPLAN Outstanding Doctoral Dissertation Award 2008*.
- Member of *UK Computing Research Committee*, since June 2008.
- Founding member of *CryptoForma*, an EPSRC Network of Excellence on *Formal Methods and Cryptography: The Next Generation of Abstractions*, 2009–2016.
- Convenor of the Microsoft Research University of Edinburgh Joint Initiative in Informatics, 2011–2018.
- Panel member for EPSRC/GCHQ Cyber Research Institute (one of panel of five deciding on £3M), June 2012
- Member of Editorial Board, Springer Series on Information Security and Cryptography, 2012–2019.
- Member of Steering Committee for ACM SIGPLAN POPL conference, 2016–2019.
- Chair of Steering Committee of *Languages for Inference*, 2019–2022.
- Member of the Executive Committee of the *UK Computing Research Committee*, 2021–2024.
- Member of the *Scientific Committee for École de Printemps d’Informatique Théorique on Probabilistic Programming (EPIT 2026)*, since October 2024.

Conference Organisation

- Co-organiser of workshop series *Higher Order Operational Techniques in Semantics (HOOTS)*, Cambridge 1995, Stanford 1997, Paris 1999, and Montreal 2000. (HOOTS book published by CUP in January 1998.)
- Convenor of *Workshop on Relations and Data Integrity Constraints and Languages (RADICAL 2010)*, Cambridge, May 2010.
- Organised *CryptoForma* workshop, March 2012, over 35 participants, from MSR (Fournet) and across UK.
- Organised Cybersecurity workshop, May 2012, with representatives of all Scottish universities.

- Co-organiser of workshop on *Algebraic Process Calculi: The First Twenty Five Years and Beyond*, Bertinoro, August 2005.
- Convenor of *Joint MSR-HP Workshop on “The Rise and Rise of the Declarative Datacentre”*, Cambridge, May 2008.
- Co-organiser of *0th CryptoForma Meeting*, Cambridge, January 2009.
- Local organiser of *Formal Methods and Tools for Security (FMATS)*, Microsoft Research Cambridge, 2013.
- Co-organiser of *Challenges and Trends in Probabilistic Programming* (Dagstuhl Seminar 15181), April 2015.
- Programme Committee Chair of POPL 2017: *44th ACM SIGPLAN Symposium on Principles of Programming Languages, Paris, January 2017*. (ACM is the leading learned society in computer science, and POPL is the premier venue for research on programming languages, a core foundation for computer science. To be PC chair is most prestigious service to the community.)
- Co-chair of *Probabilistic Programming Languages, Semantics, and Systems*, Los Angeles, January 2018.
- Co-organiser of session on *Future of Spreadsheets*, Microsoft Faculty Summit, Redmond, August 2019.

Invited Conference and Workshop Lectures

1. *MATHFIT Workshop*, London, April 1998.
2. *Security Workshop*, Gothenburg, June 1999.
3. *Workshop on Mobile Calculi*, Chennai, December 1999.
4. *Mathematical Foundations of Program Semantics (MFPS XVII)*, Hoboken, April 2000.
5. *Mathematical Foundations of Computer Science (MFCS 2000)*, Bratislava, August 2000.
6. *Applied Semantics (APPSEM)*, Darmstadt, March 2001.
7. *Automated Verification Of Critical Systems (AVoCS’01)*, Oxford, April 2001.
8. *Static Analysis Symposium (SAS’01)*, Paris, July 2001.
9. *Principles and Practice of Declarative Programming (PPDP’01)*, Firenze, September 2001.
10. *Formal Methods for Industrial Critical Systems (FMICS’02)*, Malaga, June 2002.
11. *IFIP Theoretical Computer Science (TCS 2002)*, Montreal, August 2002.

12. *Concurrency Theory (CONCUR'02)*, Brno, August 2002.
13. *Workshop on Algebraic Development Techniques (WADT'02)*, Frauenchiemsee, September 2002.
14. *Formal Aspects of Security (FASec)*, Royal Holloway College, University of London, December 2002.
15. *Workshop on Issues in the Theory of Security (WITS)*, Pisa, 2003.
16. *Foundations of Global Computing (FGC'03)*, Eindhoven, June 2003.
17. *Grids and Applied Language Theory (GALT'03)*, Edinburgh, October 2003.
18. *Formal Methods for Components and Objects (FMCO'03)*, Leiden, November 2003.
19. *UK-UbiNet Workshop*, Cambridge, May 2004.
20. *British Colloquium for Theoretical Computer Science (BCTCS)*, Nottingham, April 2005.
21. *Static Analysis Symposium (SAS'05)*, London, September 2005.
22. *Current and Emerging Research Issues in Computer Security (CERICS)*, Royal Holloway, July 2006.
23. *Logic in Computer Science (LICS'06)*, Seattle, Aug 2006.
24. *Fun in the Afternoon*, Cambridge, May 2007.
25. Panel member, *Symposium on Computer Security Foundations (CSF'07)*, Venice, July 2007.
26. *Virtual Infrastructure Workshop*, part of LISA systems administration conference, San Diego, November 2008.
27. Invited speaker, *British Colloquium for Theoretical Computer Science (BCTCS)*, Warwick, April 2009.
28. Invited speaker, *Microsoft Research Summer School*, MSR Cambridge, Cambridge, July 2009.
29. Panel member, *Symposium on Computer Security Foundations (CSF'09)*, Long Island, July 2009.
30. Invited speaker, *Workshop on Interactive Theorem Proving*, University of Cambridge, August 2009.
31. Invited lecturer, *International Summer School on Advances in Programming Languages*, Heriot-Watt University, Edinburgh, September 2009.

32. Invited speaker, *Workshop on Theory Engineering*, University of Cambridge, February 2010.
33. Invited speaker, *International Symposium on Engineering Secure Software and Systems*, Pisa, February 2010.
34. Panel member, *4th International Workshop on Analysis of Security APIs (ASA-4)*, Edinburgh, July 2010.
35. Co-organiser of special session at ACM POPL 2011 in memory of Robin Milner.
36. Invited speaker at *Microsoft Software Summit*, Paris, April 2011.
37. Invited speaker on *Strategic Thinking for Researchers*, at *Microsoft Research Summer School*, MSR Cambridge, Cambridge, July 2012.
38. Unifying Speaker at ETAPS, Eindhoven, April 2016, on *Structure and Interpretation of Probabilistic Programs*. (The European Joint Conferences on Theory and Practice of Software (ETAPS) is the primary European forum for academic and industrial researchers working on topics relating to Software Science. The unifying speakers address a plenary session of all five constituent conferences of ETAPS.)
39. Keynote speaker on *End-User Probabilistic Programming* at the *16th International Conference on Quantitative Evaluation of SysTems (QEST 2019)*, Glasgow, September 2019.
40. Invited speaker on *Project Yellow: Bringing Data Types and Functional Programming to Excel* at *JPMorgan Chase*, March 2020.
41. Invited speaker on *Excel meets Lambda* at *Lambda Days 2021* virtual conference, February 2021.
42. Invited presenter at *Excel Virtually Global 2021*, October 2021.
43. Invited speaker on *Why Statistical Thinking is Transforming Programming Language Research* at the *Colloquium on Probabilistic Programming*, Collège de France, June 2022.
44. Keynote speaker on *Bringing generative AI to the Excel grid: from research to practice* (with J. Williams) at the *European Spreadsheet Risk Interest Group Annual Conference*, London, July 2023.
45. Keynote speaker on *Requirements are all you need* at *International Conference on Functional Programming*, Milan, September 2024.
46. Invited panelist at workshop on *Theorem Proving and Machine Learning in the age of LLMs: SoA and Future Perspectives*, Heriot-Watt University, April 2025.
47. Speaker on *Vibe Coding Exposed: Why We're Desperately Dialing Dave Sands* at *Sandz Symposium*, Chalmers University, Gothenburg, May 2025.

48. Speaker on *Sharper AI with DSLs: Building apps the spec-driven way* (with M. Spönemann) at *Open Community Experience 2026 (OCX26)*, Brussels, April 2026. ([video](#), [slides](#))
49. Speaker on *The LLMbda Calculus: AI Agents, Conversations, and Information Flow* at *S-REPLS 19: Nineteenth South of England Regional Programming Languages Seminar*, University of Cambridge, July 2026.
50. Speaker on *If computing is an experimental science, who checks the experimenter?* at *LFCS 40: A Symposium on the Occasion of the 40th Anniversary of the Founding of the Laboratory for Foundations of Computer Science*, University of Edinburgh, September 2026.

Programme Committee Member

- 1995: *Functional Programming and Computer Architecture (FPCA)*.
- 1998: *Principles of Programming Languages (POPL)*, *International Conference on Functional Programming (ICFP)*, *High Level Concurrent Languages (HLCL)*, *Mathematical Foundations of Programming Semantics (MFPS)*.
- 1999: *European Symposium on Programming (ESOP)*, *Concurrency Theory (CONCUR)*, *Higher Order Operational Techniques in Semantics (HOOTS)* (chair).
- 2000: *International Colloquium on Automata, Languages, and Programming (ICALP)*, *Principles and Practice of Declarative Programming (PPDP)*, *Higher Order Operational Techniques in Semantics (HOOTS)*.
- 2001: *MEchanized Reasoning about Languages with variable bINDing (MERLIN)*.
- 2002: *Formal Methods for Open Object-based Distributed Systems (FMOODS)*, *Principles and Practice of Declarative Programming (PPDP)*, *Foundations of Wide Area Network Computing (F-WAN)*.
- 2003: *Foundations of Software Science and Computation Structures (FOSSACS)* (chair), *Verification, Model Checking, Abstract Interpretation (VMCAI)*, *Foundations of Computer Science (FCS)*, *Foundations of Global Computing (FGC)*, *Expressiveness in Concurrency (EXPRESS)*, *Formal Methods for Open Object-based Distributed Systems (FMOODS)*.
- 2004: *Foundations of Software Science and Computation Structures (FOSSACS)*, *International Colloquium on Automata, Languages, and Programming (ICALP)*, *Trust Management, Computer Security Foundations Workshop (CSFW)*, *Web Services and Formal Methods (WS-FM)*, *Formal Methods in Security Engineering (FMSE)*, *Secure Web Services (SWS)*.

- 2005: *Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, *Typed Lambda Calculi and Applications (TLCA)*, *Formal Methods for Open Object-based Distributed Systems (FMOODS)*, *International Colloquium on Automata, Languages, and Programming (ICALP)*, *Security Issues in Coordination Models, Languages, and Systems (SecCo)*.
- 2006: *International Symposium on Secure Software Engineering (ISSSE)*, *Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, *Computer Security Foundations Workshop (CSFW)*, *International Colloquium on Automata, Languages, and Programming (ICALP)*, *Formal Methods in Security Engineering (FMSE)* (chair), *Privacy Security Trust (PST)*.
- 2007: *Principles of Programming Languages (POPL)*, *Foundations of Software Science and Computation Structures (FOSSACS)*, *International Conference on Service-Oriented Computing (ICSOC)*, *Formal Methods in Security Engineering (FMSE)*.
- 2008: *International Conference on Service-Oriented Computing (ICSOC)*.
- 2009: *Practical Aspects of Declarative Languages (PADL)*, *Security and Privacy*, *Symposium on Computer Security Foundations (CSF)*, *Uk eScience All Hands Meeting (AHM)*, *ML Workshop*, *International Symposium on Fundamentals of Computation Theory (FCT)*
- 2010: *European Symposium on Programming (ESOP 2010)* (chair), *Symposium on Computer Security Foundations (CSF 2010)*, *First CryptoForma Workshop*, *Programming Languages and Analysis for Security (PLAS 2010)*.
- 2012: *Principles of Security and Trust (POST)*, *Computer Aided Verification (CAV)*, *Computer and Communications Security (CCS)*.
- 2013: *Principles of Programming Languages (POPL)*, *Principles of Security and Trust (POST)*.
- 2017: *Principles of Programming Languages (POPL)*.
- 2020: *Wikidata Workshop 2020*.
- 2021: *Wikidata Workshop 2021*, *International Conference on Functional Programming (ICFP 2021)*.
- 2022: *Wikidata Workshop 2022*
- 2023: *Object-Oriented Programming, Systems, Languages & Applications 2023 (OOPSLA 2023)*.
- 2024: *European Symposium on Programming (ESOP 2024)*.
- 2025: *15th Annual Workshop on the Intersection of HCI and PL (PLATEAU 2025)*.

- 2026: *18th International Symposium on Functional and Logic Programming (FLOPS 2026)*, *16th Annual Workshop on the Intersection of HCI and PL (PLATEAU 2026)*, *IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC 2026)*.
- 2027: *Object-Oriented Programming, Systems, Languages & Applications 2027 (OOPSLA 2027)*.

Research Grants

- Principal Investigator on EPSRC Project *An Operational Theory of Objects*, with A. Pitts and Harlequin Ltd, February 1997 to January 2000.
- Co Investigator on EPSRC Project *App Guardian: Resilient Application Stores*, October 2013 to September 2016.

Teaching and Examining

Undergraduate Teaching

- Director of Studies in Computer Science, Newnham College, Cambridge, 1994–1996.
- Undergraduate course on *Functional Programming*, University of Cambridge, spring 1995.

Graduate Courses

- *Theorem Proving Using HOL*, Chalmers University, spring 1993.
- *Bisimilarity as a Theory of Functional Programming*, Glasgow University, August 1994.
- *Bisimilarity as a Theory of Functional Programming*, University of Aarhus, March 1995.
- *Operational Methods*, University of Cambridge, winter 1995.
- *Nominal Calculi for Security and Mobility*, First International School on Foundations of Security Analysis and Design (FOSAD), Bertinoro, September 2000.
- *Security*, International School on Formal Methods for the Design of Computer, Communication and Software Systems: Process Algebras, Bertinoro, July 2001.
- *A Calculus for Cryptographic Protocols*, University of Cambridge, winter 2001.
- *A Calculus for Cryptographic Protocols*, Summer School on Foundations of Internet Security, Duszynski Zdrój, June 2002.

- *Sécurité Logicielle* (Software Security) (with C. Fournet), L'École jeunes chercheurs en programmation, Aussois, June 2003.
- *Secure Global Computing with XML Web Services: Theory and Practice*, EEFGlobal Computing Summer School, University of Edinburgh, July 2003.
- *Web Services and Security*, Fourth International School on Foundations of Security Analysis and Design (FOSAD), Bertinoro, September 2004.
- *Protecting Alice from Malice*, University of Cambridge, spring 2006.
- *Protecting Alice from Malice*, NATO Summer School Marktoberdorf, July 2006.
- *Declarative Datacentres*, GLOBAN 2008 Summer School (Global Computing Approach to Analysis of Systems), Warsaw, September 2008.
- *Principles and Applications of Refinement Types*, Winter School on Hot Topics in Distributed Computing, La Plagne, March 2009.
- *Principles and Applications of Refinement Types*, NATO Summer School Marktoberdorf, July 2009.
- *Cryptographic and Probabilistic Programming*, Fifteenth International School on Foundations of Security Analysis and Design (FOSAD), Bertinoro, September 2015.
- *Empowering Spreadsheet Users with Probabilistic Programming*, First School on Foundations of Programming and Software Systems: Probabilistic programming (ProbProgSchool 2017), Minho, June 2017.
- *Empowering Spreadsheet Users with Probabilistic Programming*, Oregon Programming Languages Summer School, Eugene, Oregon, June 2019.

External Examiner

1. M. Larsson, Licentiate, University of Linköping, August 1993.
2. J. Ross, PhD, University of Cambridge, August 1997.
3. M. Norrish, PhD, University of Cambridge, November 1998.
4. C. Taylor, PhD, University of Nottingham, October 1998.
5. J. Kleist, PhD, University of Aalborg, March 2000.
6. A. Unyapoth, PhD, University of Cambridge, April 2001.
7. M. Loretto, PhD, University of Florence, January 2002.
8. S. Crafa, PhD, University of Venice, December 2002.
9. M. Grazia Vigliotti, PhD, Imperial College, June 2004.

10. M. Becker, PhD, University of Cambridge, September 2005.
11. M. Maffei, PhD, University of Venice, January 2006.
12. A. Phillips, PhD, Imperial College, January 2006.
13. N. Broberg, Licentiate, Chalmers University, September 2006.
14. S. Tse, PhD, University of Pennsylvania, February 2007.
15. J. Borgström, PhD, EPFL, January 2008.
16. P. Cerny, PhD, University of Pennsylvania, May 2009.
17. E. Cooper, PhD, University of Edinburgh, September 2009.
18. A. Pironti, PhD, University of Turin, February 2010.
19. J. Jakubuv, PhD, Heriot-Watt, 2010.
20. L. Hu, PhD, University of Nottingham, February 2011.
21. B. Smyth, PhD, University of Birmingham, March 2011.
22. C. Hrițcu, PhD, Saarland University, January 2012.
23. T. Given-Wilson, PhD, Sydney University of Technology, 2012.
24. F. Gretz, PhD, MacQuarie University, 2015.
25. P. Koch, PhD, University of Klagenfurt, December 2019.
26. A. Munk, PhD, University of British Columbia, June 2023.

PhD Students

- PhD graduate, P. Hankin, *A Study of Objects*, 2001.
- PhD candidate, I. Baltopoulos, *Enriching a Multi-Tier Programming Language: Security, Concurrency, and Typing*, 2005–2009.
- PhD graduate, F. Dupressoir, *Proving Cryptographic C Programs Secure with General-Purpose Verification Tools*, 2013.
- PhD graduate, M. Aizatulin, *Verifying cryptographic security implementations in C using automated model extraction*, 2015.
- PhD graduate, M. Szymczak, *Programming Language Semantics as a Foundation for Bayesian Inference*, 2017.
- PhD graduate, M. Gorinova, *Program Analysis of Probabilistic Programs*, 2021.
- PhD graduate, E. Vlassi-Pandi, *Natural Type Inference*, 2023.
- PhD candidate, X. Sun, *Multi-Agent Systems for Traceable Bayesian Workflow*, since 2024.